A sum whose limits sit close to the lines of text around it:

$$\sum_{l=0}^{\infty} \gamma^l \tilde{r}_t = \sum_{l=0}^{\infty} \gamma^l r_{t+l} \tag{1}$$

Letting the transformed values be as above, one obtains the bound below.

$$W(\mathbb{P}_\theta, \mathbb{P}_{\theta'}) \leq \|g_\theta(z) - g_{\theta'}(z)\|$$

A script of a subscript:

$$x_i^m = x_i^{m-1} + \epsilon_i s_{\theta^*}(x_i^{m-1})$$

Accents on a symbol, not on its script, and a radical over scripts:

$$\tilde{f}_i(x) = d\bar{w} + \sqrt{\sigma_i^2 - \sigma_{i-1}^2}$$

A tall bracket built from pieces:

$$y = \begin{bmatrix} a \\ b \\ c \\ d \\ e \\ f \end{bmatrix}$$